

Kestrel Spectrograph: UV Channel Layout

1 Description

This note describes the optical layout of the Kestrel spectrograph's ultraviolet channel, from the beam it receives to its detector, and the guide camera that shares its input.

2 Arrangement

Light enters at the input beam, passes the fold mirror FM1, the dichroic D1, OAP1 in turn, and reaches the field stop. Downstream of the field stop comes the ellipsoidal mirror EM1. Downstream of the ellipsoidal mirror EM1 comes the filter wheel. Downstream of the filter wheel comes the grism G1. Downstream of the grism G1 comes EMCCD. The guide camera receives the light from the dichroic D1 in guide mode. The field stop sits at a focus. The filter wheel sits at a pupil. The grism G1 sits at a pupil. The filter wheel can be selectable as needed. The grism G1 can be selectable as needed. UV channel houses OAP1. The field stop sits within UV channel. UV channel houses the ellipsoidal mirror EM1. The filter wheel sits within UV channel. The grism G1 sits within UV channel. UV channel houses EMCCD. Either the wide stop or the narrow stop can be selected for the field stop.